

Education

Georgia Institute of Technology – Atlanta, Georgia (Expected graduation date December 2026)
Master of Science in Computer Science

George Mason University – Fairfax, Virginia
Bachelor of Science in Computer Science

Relevant coursework: Artificial Intelligence, Deep Learning, Machine Learning, Computer Vision, Machine Learning for Trading, Operating Systems, Secure Programming, Data Structures, Analysis of Algorithms

Skills

Programming: Python, Java, C, C++, C#, Object-Oriented Programming, Agile, SQL, REST, Spring Boot, .NET Framework

Machine Learning/AI: NumPy, SciPy, scikit-learn, PyTorch, OpenCV

Applications: Git, Unity, Microsoft Office, Adobe Creative Suite, Apache Tomcat, Oracle, TeamDynamix

Other Skills: Windows, Unix, OpenSSL, GDB, Unit Testing, Technical and Legal Research, Analysis, Executive Decision Making

Related Projects

Optimized Supervised and Unsupervised Learning Project September – November 2025

- Developed and evaluated supervised learning models (decision trees, kNN, SVM, neural networks) on real-world datasets using rigorous validation and leakage controls.
- Conducted optimizer and randomized optimization ablation studies, and evaluated regularization techniques.
- Analyzed model capacity and generalization by comparing architectures, kernels, and optimization strategies.
- Applied clustering and dimensionality reduction to uncover latent structure and enhance feature representations.
- Integrated reduced-dimension and cluster-derived features into neural networks and produced empirical diagnostics.
- Authored comprehensive technical reports documenting methodology, experiments, and data-driven findings.

Malicious Network Traffic Detection Project July 2025

- Worked with a team to develop a deep learning-based Network Intrusion Detection System (NIDS) to identify cyberattacks and anomalous network behavior without reliance on signature-based detection.
- Implemented and benchmarked multiple architectures (DNN, CNN, RNN, and LSTM) to evaluate effectiveness.
- Utilized the CSE-CIC-IDS2018 dataset to train and validate models against realistic attack patterns.
- Compared model performance using metrics such as F1 score, classification accuracy, and precision.
- Leveraged Python-based deep learning frameworks and utilities to implement scalable training and evaluation pipelines.

Machine Learning Stock Trading Project April 2024

- Engineered an AI strategy learner that autonomously developed trading rules using machine learning techniques, specifically utilizing Random Forest and Bag Learning algorithms to adapt and predict market movements.
 - Leveraged financial indicators to create actionable trading signals for stock market operations.
 - Conducted rigorous back-testing of both strategies to compare their effectiveness against a benchmark portfolio.
 - Designed and executed experiments to fine-tune parameters and assess the impact of market conditions on performance.
 - Authored a detailed analytical report summarizing trading strategy outcomes, methodology, and experimental results.
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Work Experience

Virginia Supreme Court – Fairfax, VA

Fairfax County Magistrate, July 2024 – Current

- Adjudicate initial hearings in both criminal and civil cases, issuing rulings that enforce local and state laws.
- Issue various types of legal orders such as arrest warrants, search warrants, and emergency protective orders based on detailed evaluation of affidavits and sworn testimonies.
- Conduct bail hearings to assess the risk factors and determine appropriate bail conditions for defendants.
- Analyze and interpret statutory requirements and judicial precedents to ensure fair and lawful proceedings.
- Uphold ethical standards and impartiality, demonstrating integrity and fairness in all judicial decisions.

George Mason University – Fairfax, VA

Classroom Technologies Assistant, Aug 2021 – Dec 2023

- Provided technical assistance and training to professors and other customers at client request
- Worked with a team of technicians to ensure the continued operation of 300 learning spaces
- Logged incidents and services daily with workflow management software TeamDynamix
- Solved technical problems over the phone and in person